1. Introduction to Quantum Information Science

Quantum computing represents a paradigm shift from classical information theory. Classical computers encode information in binary digits, or bits, which assume a deterministic value of either 0 or 1 at any given computational instant.

In stark contrast, quantum computers leverage the quantum mechanical principle of state superposition. A quantum bit, or qubit, exists in a two-dimensional complex Hilbert space. The state of a qubit is mathematically represented as:
|psi> = alpha |0> + beta |1>

where alpha and beta are complex probability amplitudes satisfying the normalization constraint |alpha|^2 + |beta|^2 = 1. The quantities |alpha|^2 and |beta|^2 dictate the probability of observing states |0> and |1> respectively upon measurement in the computational Z-basis.

This linear superposition allows an n-qubit register to simultaneously span a 2^n dimensional state space, providing an exponential representational capacity unattainable in classical systems.

2. Quantum Entanglement and Bell States

Entanglement is a purely quantum mechanical phenomenon wherein the collective quantum state of multiple particles cannot be decomposed into a tensor product of their individual single-particle states.

Mathematically, for a bipartite system A and B, a state |psi_AB> is entangled if:
|psi_AB> != |phi_A> (x) |phi_B>

The canonical maximally entangled two-qubit states are known as the four Bell states or EPR pairs:
|Phi+> = (|00> + |11>) / sqrt(2)
|Phi-> = (|00> - |11>) / sqrt(2)
|Psi+> = (|01> + |10>) / sqrt(2)
|Psi-> = (|01> - |10>) / sqrt(2)

Entangled qubits exhibit nonlocal quantum correlations that violate Bell's inequalities (CHSH inequality threshold exceeding the classical bound of 2 up to Tsirelson's bound of 2*sqrt(2)). Entanglement serves as the fundamental resource for quantum teleportation, superdense coding, and distributed quantum key distribution protocols.

3. Quantum Logic Gates and Reversible Computation

Because time evolution in closed quantum systems is governed by the Schrödinger equation, all quantum operations must be unitary and therefore strictly reversible. A quantum gate acting on n qubits corresponds to a 2^n x 2^n unitary matrix U such that U^dagger U = I.

Crucial Single-Qubit Gates:
- The Pauli-X Gate (bit-flip): Maps |0> to |1> and |1> to |0>. Matrix: [[0, 1], [1, 0]].
- The Pauli-Z Gate (phase-flip): Leaves |0> unchanged and maps |1> to -|1>. Matrix: [[1, 0], [0, -1]].
- The Hadamard Gate (H): Creates an equal superposition from computational basis states. H|0> = (|0> + |1>) / sqrt(2), and H|1> = (|0> - |1>) / sqrt(2).

Crucial Multi-Qubit Gates:
- Controlled-NOT (CNOT) Gate: Flips the target qubit if and only if the control qubit is in state |1>. When combined with arbitrary single-qubit rotations, the CNOT gate forms a universal set of quantum gates for quantum computation.

4. Quantum Teleportation Protocol

Quantum teleportation enables the transmission of an unknown quantum state |psi> = alpha |0> + beta |1> from sender Alice to receiver Bob without physically transferring the underlying physical particle, provided Alice and Bob share an entangled Bell pair |Phi+>.

Protocol Execution Steps:
Step 1: Alice holds the unknown qubit |psi> and one half of the shared Bell pair. Bob holds the other half.
Step 2: Alice applies a CNOT gate between |psi> and her Bell qubit, followed by a Hadamard gate on |psi>.
Step 3: Alice performs a projective Bell-basis measurement on her two qubits, yielding one of four classical bit outcomes: 00, 01, 10, or 11.
Step 4: Alice transmits the two classical measurement bits to Bob via a classical channel.
Step 5: Based on the received classical bits, Bob applies a corrective Pauli operator (I, X, Z, or ZX) to reconstruct state |psi> on his qubit.

Crucially, quantum teleportation does not violate the relativistic speed of light because the classical transmission of two bits is strictly necessary to reconstruct the quantum state.

5. Foundational Quantum Algorithms

Several algorithms demonstrate verifiable quantum advantages over optimal classical algorithms:

- Shor's Factoring Algorithm: Solves the integer factorization problem in polynomial time O((log N)^3) by reducing factoring to order-finding using the Quantum Fourier Transform (QFT). This represents an exponential speedup over the best known classical algorithm, the General Number Field Sieve.

- Grover's Search Algorithm: Solves the unstructured search problem over an unsorted database of N entries in O(sqrt(N)) evaluations of the oracle function, representing a quadratic speedup over classical brute-force search O(N).

- Quantum Phase Estimation (QPE): Approximates the eigenvalue corresponding to an eigenvector of a unitary matrix, forming the underlying subroutine in Shor's algorithm and Hamiltonian simulation for quantum chemistry.